

Nagare — Daily Progress Report

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For: Prof. Kato (NITech, Katolab) and Katalin — collaboration update

Repository: <https://github.com/kyberszittya/nagare>

Summary

Today Nagare progressed from an in-framework component to a **standalone, machine-verified, and benchmarked** closed-form learning framework, with a first **competitive result on a real signed-link-prediction task**. The headline: on a fair CPU-vs-CPU comparison Nagare matches or beats PyTorch while using far less memory; its balance/holonomy foundation is now formally verified; and its closed-form learner reaches competitive AUROC on Slashdot/Epinions/Bitcoin. We also report, honestly, the one differentiating experiment that remains open.

1. Framework consolidation

Nagare was extracted into a self-contained public repository: the two diverged copies were reconciled, the framework dependencies (`hymeko_clifford`, `hymeko_graph`) were vendored in, and the crate now builds with no coupling to the parent framework. It compiles and runs on both Windows and the Katolab GPU host (`kato15`, Linux); **45 tests pass** on both. All optimisations below are proven bit-identical (finite-difference tests + a frozen fixture).

2. Performance — CPU, versus PyTorch

Nagare replaces the autograd tape with contiguous, auto-vectorised, closed-form kernels that parallelise (rayon) to every core. Two profile-driven fixes (a loop reorder that enables SIMD, and parallelising the pooling stage) took the forward from behind PyTorch to ahead of it on CPU.

axis	metric	Nagare (CPU)	PyTorch (CPU)
forward latency (toy shape, 8 threads)	$\mu\text{s}/\text{sample}$	7.6	7.6–9.1
dimensionality sweep (best-of-each)	speed-up	1.2–2.6× faster	baseline
peak memory (toy → scaled)	RSS ratio	3–160× less (Nagare)	

Honest bounds: at *matched* 8 threads on large dense matmuls, tuned BLAS (MKL) still wins; and Nagare has *no GPU backend yet*, so a GPU is faster in absolute terms — that is future work, not a CPU-vs-GPU verdict. The genuine, robust win is competitive CPU speed at a large memory advantage.

3. Theoretical foundation — machine-verified

Nagare’s learning target is *signed balance* = $\mathbb{Z}_2$ *holonomy*. We machine-verified the load-bearing statements with an SMT solver (Z3) and symbolic algebra (sympy): the Cartwright–Harary balance theorem (switchable $\iff$ every cycle positive), the gauge invariance of cycle holonomy, and the cycle-space structure — each proved as a validity, not sampled. Empirically, real signed networks are strongly balanced (87–93% **balanced triads**), which is the structural reason holonomy is predictive.

4. Signed-link prediction — first real-task result

A pure-Nagare closed-form model (local update rule, no backpropagation) on holonomy features achieves, leakage-free, 5-datasets, standard 80/20 protocol:

	Bitcoin-Alpha	Bitcoin-OTC	Slashdot	Epinions
Nagare (closed-form) AUROC	0.904	0.928	0.910	0.951

This is **competitive** (inside the published SGCN/SiGAT band ~ 0.93 – 0.97) — Nagare’s closed-form learner *matches* the baselines. It does not beat them, because on these datasets the signal is triad-saturated (deeper cycles and continuous edge weights add little to the *binary* sign target). The value is therefore competitive quality via closed-form learning at the speed/memory advantage above, not an accuracy breakout.

5. Open question / next step

Signed hypergraphs are naturally real-valued ($w_e \in [-1, 1]$), and Nagare’s Clifford/rotor holonomy models that continuum. The measurements show its payoff is *not* on binary sign prediction but on a **continuous target** — predicting rating / trust *strength* (regression), where the magnitude is necessary and binarising cannot compete. That is the single, well-posed differentiating experiment remaining.

Artifacts (on the repository)

Integrated technical draft `docs/nagare-paper.pdf`; reports and figures under `reports/`; reproducible harnesses under `examples/` and `scripts/dev/` (theorem verifier, benchmarks, signed-link model).

Prepared as an honest status update: measured results are labelled as such, and the one unproven differentiating claim is stated as open.